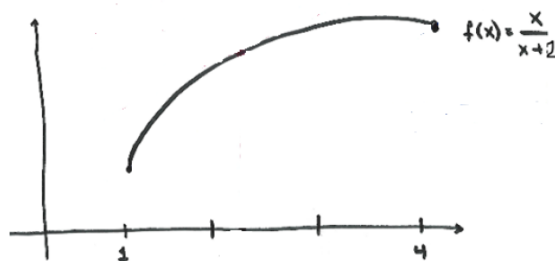


Problem 4

Problem 4

Problem 1 Verify that the given function satisfies the hypotheses of the Mean Value Theorem in the given interval. Then algebraically find all numbers c that satisfy the conclusion of the Mean Value Theorem. Using the graph provided, label the point(s) c and sketch the secant line through the points $(1, f(1))$ and $(4, f(4))$ and the tangent line at c .

$$f(x) = \frac{x}{x+2} \quad \text{on } [1, 4]$$



Explanation. The function $f(x) = \frac{x}{x+2}$ is continuous on $[1, 4]$ and differentiable on $(1, 4)$ since it is a rational function, and is therefore continuous and differentiable on its domain. Therefore, f satisfies the hypotheses of the Mean Value Theorem. We have that

$$f'(x) = \frac{(x+2)(1) - x(1)}{(x+2)^2} = \frac{2}{(x+2)^2}$$

$$\frac{f(4) - f(1)}{4 - 1} = \frac{\frac{2}{3} - \frac{1}{3}}{3} = \frac{1}{9}$$

So we are looking to find all points $c \in (1, 4)$ which satisfy that $f'(c) = \frac{1}{9}$. So we solve:

$$\frac{2}{(c+2)^2} = \frac{1}{9}$$

$$(c+2)^2 = 18$$

$$c = \sqrt{18} - 2 = 3\sqrt{2} - 2$$

Note that we omitted $-\sqrt{18} - 2$ above because it is not in the interval $(1, 4)$. Therefore, $c = 3\sqrt{2} - 2$.

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